

## Green AMD

To read more on green initiatives by AMD and other IT companies, read the April 2010 issue of Digit

## Thinner Solar Panels

Oregon State University have unveiled nanostructure solar films, that will considerably reduce material waste and cost of solar panels

Graphics Card on the Planet' title, with the HD 5970, and that title still has not been taken up by the competition even after the six month delay.

We are actually in a strong leadership position. We are not really threatened by the Fermi as a product, because of the strategy that we incorporated with the Evergreen products, where the mid-range solution is designed first before being scaled up to performance range and scaled down to the value range, because we have the advantage of manufacturing, and no great complexities involved. What the competition has adopted is that they build a huge massive die, which is really tough to manufacture, and really power hungry.

The specifications announced when the Fermi was introduced are not the same as what they [Nvidia] are releasing today. So, that was a strategy that really kept us going in the graphics market, and today we have the best graphics products in the world.

### When approximately will the Opteron 4100 series processors be launched? Will they also work on Socket G34?

They will be launched in the next quarter, and no, they will not fit on the Socket G34. The reason why we are launching the Opteron 4100 series is to bring the volume servers to market. The socket they are designed to fit on is C32 socket, which will be common to the 1P and 2P market. The G34 socket will be common to the 2P and 4P market.

The Opteron 4100 is targeted for entry-level stacks or Rack Density Deployments. Someone who is really, really bothered about power, and want very really-low power consuming products, would go for the 4100 series, especially in applications like cloud, where one would want a very rapid, dense deployment. When one is talking about cloud, one is really not bothered about how many memory channels you have, how much memory each processor has, but instead, one is bothered about what is the computing resources available are. Therefore, it is specifically targeted for those kind of rack density deployments.

### The upcoming Xeon 7500 Nehalem EX top-end processor, the X7560, will have almost twice the shared L3 cache, 4 more threads per processor, and 4 less cores per processor than the top-end 6176 SE Magny Cours. How do AMD's Opteron 6100 processors rack up against this and still provide competitive performance?

Both are targeted at different customers with different requirements. The Nehalem EX comes with all the fancy specifications that you just talked about, but it was meant for a different class of customers who are looking for risk-free placements. The cost becomes the biggest apples to oranges comparison factor there. The Magny Cours are



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meant for customers who want to go for value 4P servers, who want to keep their data centres at a cost-conscious level. Both are different products to be compared. You can almost buy 2 4P Magny Cours parts at the cost of approximately 1 Nehalem-EX processor. That's the biggest advantage.

There are also a whole lot of other challenges that Intel will have with the Nehalem-EX. The introduction of very expensive memory buffers on the motherboard, which will make the overall cost of the server unaffordable to IT managers, especially at this time, when the market is recovering from a beating, and when IT managers are very cost-conscious. A 4P AMD server will meet his requirements.

### Is there any significant performance gain other than scalability in maintaining the same chipset architecture across all platforms unlike Intel?

There is a huge advantage in having common chipset architecture between 1P, 2P and 4P. The first advantage is drivers. Develop drivers once, and use it all across. So, the cost of development comes down for our OEMs. Second advantage is, 1P and 2P are volume servers, and the low cost of chipset manufacturing will be suitable for them, and that economics can be rolled for 4P. 4P has higher manageability and higher reliability factor, so when you develop a driver for that particular solution, you can use it for 4P. So, one segment has an advantage using another segment's solutions. The same thing applies for BIOS code, develop it for one solution, and use it across designs.

### We have seen that 8-core and 12-core AMD architecture is targeted on server platforms. However, is it optimized for home applications like multimedia and gaming?

Typically, the server receives the technology first, before they are brought into home use. Well, if someone wants to use a 12-core Magny Cours processor on their desktop for gaming, they can. There's no problem, but the laws of economics do not work there, the cost and power consumption will not be feasible for gaming. The Opteron Series is therefore for enterprise markets, based on the demands and needs of data centres and servers.

### What is your dream gaming PC configuration?

A gaming PC, well that's our strength – graphics are our strength. What I use for gaming in my lab is the AMD Phenom II X4 965 Black Edition which I have overclocked to 4.2GHz from 3.4GHz, and I am really happy with its performance. What else is on the machine is our flagship product, the Evergreen HD 5970, which I have applied in a Crossfire mode and Eyefinity, and the power supply is close to 1500 watts. So, I do have the fastest gaming machine in India to be precise, and I really enjoy it. 